

EARLY ELECTRONICS



**TEACHER:
CERTIFIED
TRAINER**

**DURATION:
16 CLASSES (1 HOUR PER
CLASS)**

**MODE:
OFFLINE**

This is a powered, hands-on and snap-based electronics learning solution designed for early learners. Students use safe, modular snap-fit circuit components to build, assemble and test working electronic circuits, helping them connect textbook concepts to real-world applications. Each session involves creating functional circuits such as lights, buzzers, sensors and control systems, allowing students to clearly understand how electronic systems work in everyday devices. Through guided experiments and activities, learners explore core STEM concepts including current flow, voltage, resistance, switches, sensors and basic logic in a simple and engaging way. High-quality, child-safe snap-based electronic kits made from durable materials are used to ensure reliable, reusable and safe learning experiences while encouraging curiosity, creativity and problem-solving skills.

COURSE CURRICULUM

Early Electronics – Level 1

1. Introduction to Electronics

- Introduction to electronics and its role in daily life.
- Understanding electricity, electronic components and simple circuits.

2. Series and Parallel Circuits

- Understanding series circuits and how current flows in one path.
- Understanding parallel circuits and multiple current paths.
- Comparing brightness, power flow and real-life examples.

3. Combinational Circuits and Related Projects

- Combining series and parallel circuits.
- Building simple working models using snap-based components.
- Understanding practical applications of combinational circuits.

4. Smart Watch Display

- Understanding basic display concepts.
- Creating a simple display-style circuit to simulate a smart watch interface.

5. Logic Gates

- Introduction to logic gates (AND, OR, NOT).
- Understanding inputs, outputs and basic logic operations using simple examples.

6. Project Based on Logic Gates

- Building small projects using logic gates.
- Learning decision-making circuits used in real-life electronics.

7. Introduction to Integrated Circuits (ICs)

- Understanding what ICs are and why they are used.
- Identifying IC pins and basic functions.

8. IC-Based Projects

- Creating simple circuits using ICs.
- Learning how ICs simplify complex electronic systems.

9. Introduction to Potentiometer

- Understanding variable resistance.
- Learning how potentiometers control volume, brightness and speed.

10. Introduction to Light Dependent Resistor (LDR)

- Understanding how LDR responds to light.
- Learning day–night and automatic light concepts.

11. Projects on LDR

- Building light–sensing circuits.
- Real-life applications such as automatic street lights.

12. Introduction to Capacitor

- Understanding energy storage in capacitors.
- Learning charging and discharging concepts.

13. Project Day – 1

- Guided project using learned components.
- Focus on creativity and correct circuit building.

14. Project Day – 2

- More advanced project implementation.
- Improving troubleshooting and testing skills.

15. Project Day – 3

- Independent project completion.
- Demonstration and explanation of working models.

16. Challenge Day

- Problem-solving challenges using snap-based circuits.
- Testing understanding, speed and creativity.

Early Electronics – Level 2

1. PNP Transistors

- Understanding transistor basics.
- Learning PNP transistor working and applications.

2. NPN Transistors

- Understanding NPN transistor operation.
- Comparing NPN and PNP transistors with examples.

3. Recording IC

- Introduction to sound recording and playback ICs.
- Understanding how voice recording circuits work.

4. Amplifier

- Understanding signal amplification.
- Learning how weak signals are made stronger.

5. SCR (Silicon Controlled Rectifier)

- Understanding SCR as an electronic switch.
- Learning its use in control and protection circuits.

6. Project Based on SCR

- Building control-based circuits using SCR.
- Understanding real-life applications like alarms and light control.

7. Relay

- Understanding relay as an electrically operated switch.
- Learning how low-power circuits control high-power devices.

8. Transformer

- Understanding voltage step-up and step-down concepts.
- Learning safe power transfer principles.

9. FM Module

- Introduction to wireless communication.
- Understanding signal transmission and reception.

10. Variable Capacitor

- Understanding capacitance variation.
- Learning tuning concepts used in radios.

11. Solar Cell

- Understanding solar energy conversion.
- Learning renewable energy basics through experiments.

12. Antenna Coil

- Understanding electromagnetic waves.
- Learning the role of antenna coils in communication systems.

13. Project Day – 1

- Guided advanced electronics project.
- Focus on correct connections and testing.

14. Project Day – 2

- Intermediate-level project building.
- Improving circuit analysis and debugging skills.

15. Project Day – 3

- Final project implementation.
- Presentation and working demonstration.

16. Challenge Day

- Advanced electronics challenges.
- Application-based problem solving and teamwork.

Class Structure

1. Introduction to Electronic Components

- Learn about basic and advanced electronic components and their functions.

2. Snap-Based Circuit Assembly

- Build circuits using safe, modular snap-fit electronic kits under supervision.

3. Concept Understanding Through Practice

- Simple explanations of electronic principles linked to real-life devices.

4. Hands-on Activities and Projects

- Interactive projects, experiments and challenges to reinforce learning.

5. Safety, Discipline and Real-Life Applications

- Learning safe handling of electronic components and understanding their use in everyday technology.